Test-Time Attention Purification for Backdoored Large Vision Language Models

Appendix

We summarize the Appendix as follows:

- Appendix A provides a detailed setting of the adopted models (Appendix A.1), datasets (Appendix A.2), backdoor attacks (Appendix A.3), and backdoored input purification baselines (Appendix A.4).
- Appendix B provides detailed guidelines on how to select cross-modal fusion layers.

A. Detailed Settings

A.1. Models

- **LLaVA-1.5.** [9] is a popular open-source LVLM that integrates a CLIP ViT-L/336 visual encoder with the Vicuna LLM. It improves upon the original LLaVA by adopting stronger training data and an enhanced instruction-following pipeline, enabling robust performance on captioning, VQA, and general multimodal reasoning tasks. LLaVA-1.5 is widely used in prior works on LVLM robustness and backdoor research, making it a standard backbone for fair comparison.
- **Qwen2-VL** [1] is a next-generation vision–language model from the Qwen2 family, featuring a high-resolution visual encoder and a strong multilingual LLM backbone. It supports fine-grained visual grounding, captioning, OCR, and multi-image reasoning, and achieves state-of-the-art performance across numerous multimodal benchmarks. Compared with earlier LVLMs, Qwen2-VL exhibits stronger visual fidelity and more precise cross-modal alignment, providing a challenging and modern testbed for studying backdoor behaviors.
- **InstructBLIP** [16] builds on the BLIP-2 framework by combining a pretrained ViT-based visual encoder, a Q-former for visual token extraction, and an instruction-tuned Vicuna LLM. By aligning image features with language instructions, InstructBLIP significantly improves general vision–language instruction following versus earlier BLIP-style models. Its modular architecture separates the vision encoder, Q-former, and LLM, which offers a complementary structure for analyzing backdoor injection and defense mechanisms.

A.2. Datasets

- **VQAv2** [3] contains about 204K images and 1.1M human-authored questions, each paired with 10 crowd-sourced answers. Compared with VQAv1, VQAv2 reduces language priors by collecting complementary image–question pairs that yield different answers, making vision signals more important for answering. We follow the standard open-ended setting and report accuracy using the official evaluation protocol.
- **OK-VQA** [12] is a knowledge-based VQA benchmark also built on COCO images. It provides 14,055 open-ended questions that *require* external world knowledge beyond the image (e.g., commonsense, factual knowledge), and each question has 5 ground-truth answers. All questions are manually filtered to ensure that the image alone is insufficient, leading to a challenging setting where LVLMs must integrate visual understanding with external or parametric knowledge. We adopt the standard train/val splits.
- **MSCOCO** (Microsoft Common Objects in Context) [8] is a widely used dataset for detection, segmentation, and captioning. It contains roughly 330K images (over 200K labeled), 1.5M object instances, and 80 object categories, with 5 human-written captions per image describing everyday scenes. For captioning experiments, we follow the common 2017 split with 118K training images and 5K validation images. MSCOCO serves as our main large-scale benchmark for evaluating image-to-text generation and backdoor behavior under diverse real-world scenes. The prompt for captioning MSCOCO is "Please describe this image in a short sentence.".
- **Flickr8k** [5] is a smaller but well-established captioning dataset consisting of approximately 8,000 images, each paired with five descriptive captions written by human annotators. The images mainly depict people and animals in everyday activities, and the standard split contains 6K/1K/1K images for train/val/test. Due to its modest size, Flickr8k is commonly used for fast prototyping and ablation studies; in our work, we use it to analyze backdoor behavior and defenses under a low-data captioning regime. The prompt for captioning Flickr8k is "Please describe this image in a short sentence.".

A.3. Backdoor Attacks

- **BadNet** [4] is a classical backdoor attack that embeds a small patch into images and relabels them to the target class. Following standard practice, we adopt a patch size of 30 pixels and place the patch on the upper-left location of the images .

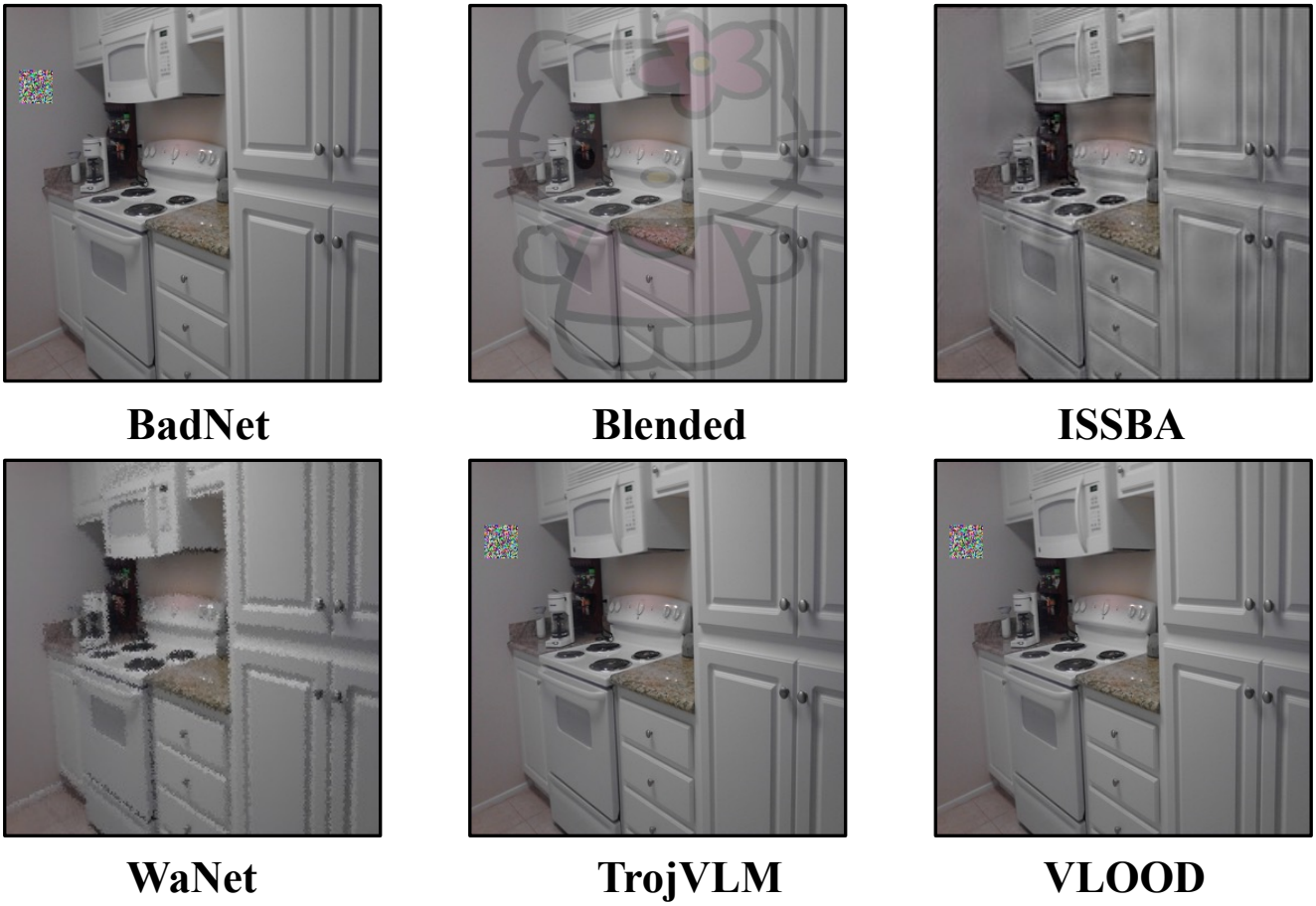

Figure 1. The visualization of the experimented backdoor attacks.

- **Blended** [2] improves stealth by linearly blending the trigger with the clean image, producing barely perceptible perturbations. We use a blending ratio of 0.2 and choose the hello kitty as the trigger image.
- **ISSBA** [7] achieves high stealthiness by generating *instance-specific* triggers. An encoder–decoder network produces a trigger encoding a predefined ciphertext, which is then embedded into each benign image. In our implementation, the encoded string is "`Stega!!`".
- **WaNet** [13] uses warping-based triggers to create smooth, imperceptible geometric distortions. We follow the standard configuration with control grid size $k = 224$ and warping strength $s = 1$, and we disable the noise mode during training.
- **TrojVLM** [10] is a backdoor attack tailored for LVLMs, designed to inject predetermined target text into the generated caption while largely preserving the original semantic content, posing a significant threat to vision–language systems. We use patch-based attack as trigger to activate TrojVLM.
- **VLOOD** [11] is an OOD-based backdoor attack for LVLMs that leverages external datasets to craft poisoned samples without requiring access to the original training set. We follow prior work and insert a fixed target phrase to implement the attack, while ensuring minimal semantic distortion. We use patch-based attack as trigger to activate VLOOD.

The visual illustration of the attcked method is shown in Figure 1.

A.4. Backdoored Input Purification Baselines

- **Blur** is implemented following a simple low-pass filtering principle. Given an input image x, we apply a PIL-based `GaussianBlur` operator with an adaptive blur radius $r = \text{intensity} \times 0.06 \cdot \min(H, W)$, where H and W are the spatial dimensions. We then linearly blend the blurred image with the original one:

$$x' = (1 - \text{intensity}) \cdot x + \text{intensity} \cdot \text{Blur}(x, r).$$

In our experiments we set intensity $= 0.7$, which corresponds to a moderately strong smoothing effect that suppresses localized trigger patterns while preserving most global semantics. This implementation directly matches the code in our release (`gauss_blur_defense`).

- **ST defense (Spatial Transform defense)** [6] is implemented as combination of multiple spatial transform techniques. For a given intensity level, we apply a sequence of stochastic geometric operations to the input image: (1) horizontal mirroring, (2) scale-down (shrink) followed by center padding back to the original resolution, and (3) random rotation with an angle sampled uniformly from $\pm 180^\circ \cdot$ intensity. All operations preserve the output size and use a neutral fill color. With intensity $= 0.7$, the transformations introduce strong yet semantics-preserving perturbations that break fine-grained spatial patterns typically relied upon by backdoor triggers. This corresponds exactly to our implementation in `_spatial_transform_pil` and `rethinking_trigger_augment`.
- **BDMAE** [15] is a test-time backdoor defense method that uses a Masked AutoEncoder (MAE) to detect and mask potential local triggers in images, and then fuses MAE restorations to reconstruct images and recover correct labels. To adapt BDMAE from image classification models to large vision–language models (LVLMs), we only retain the structural-similarity-based component. We use the same set of hyperparameters as in the original paper, which the authors have shown to be effective across multiple datasets. Specifically, we set the MAE masking ratio to 75% to occlude most patches while preserving semantics, use $N_o = N_i = 5$ random masking rounds when estimating structural-similarity-based trigger scores, and apply adaptive thresholds $\{0.6, 0.55, 0.5, 0.45, 0.4\}$ on the resulting score map to decide which patches to mask and restore. Our implementation is based on the official BDMAE code.
- **SampDetox** [17] is a black-box backdoor defense that employs a two-stage perturbation and DDPM-based denoising pipeline: it first adds lightweight global noise to suppress low-visibility triggers, and then uses structural similarity to localize and aggressively perturb visible trigger regions, enabling diffusion models to remove diverse backdoor patterns while preserving the original sample semantics. In our experiments, we strictly follow the implementation details of the original paper, including both the core code logic and diffusion model settings (i.e., $\bar{t}_1 = 20$ and $\bar{t}_2 = 120$, as recommended by the authors to control the noise intensity and the number of denoising steps). Our implementation is based on the official code and guidelines in the original paper of SampDetox.
- **ZIP** [14] is an input-purification method that first applies simple linear transformations (e.g., blurring and grayscaling, as in the original paper) to destroy backdoor patterns, and then leverages a pre-trained diffusion model to recover the semantic information removed by these transformations. In our implementation, we use a blur kernel size of 8 and set $\lambda = 5$ to empirically balance backdoor removal and semantic preservation. The code can be accessed in ZIP's github repository.

B. Guidelines of Locating the Cross-modal Fusion Start Layer.

To determine the starting detection layer ℓ_s used in our method, we follow the experimental methodology proposed by Zhang et al. [18], who systematically analyzed the internal information exchange between vision and language modalities in LVLMs. Their study investigates *where* and *how strongly* cross-modal fusion occurs across transformer layers, using a combination of controlled interventions and diagnostic probing. The procedure can be summarized as follows.

1. **Task setup.** The authors conduct their analysis on standard multimodal reasoning tasks, primarily visual question answering (VQA), which naturally requires cross-modal understanding between an input image and a textual question. For each LVLM, the model is run in a normal auto-regressive inference mode to generate answers token by token. This setup allows them to trace information flow between visual and linguistic tokens at different decoder depths.
2. **Attention knock-out intervention.** To probe which layers actually perform cross-modal fusion, Zhang et al. [18] introduce a fine-grained "attention knock-out" experiment. During inference, they manually zero out specific blocks of the attention matrix within a given transformer layer—effectively disabling attention from one modality to another—while keeping all other components intact. Three types of attention connections are tested:
   - From image tokens to question tokens (image→question edges),
   - From salient image patches to question tokens (object→question edges),
   - From image or question tokens to the answer token (fusion→answer edges).

   After each intervention, they record the degradation in model accuracy or answer likelihood. The intuition is simple: if blocking certain cross-modal attention heads in a given layer causes a substantial drop in performance, that layer must play an important role in integrating visual and linguistic information.
3. **Layer-wise information flow curves.** By repeating the above intervention for each transformer layer, they obtain a layer-resolved curve describing how much the visual stream contributes to linguistic representations. This is often measured using *cross-modal attention flow* (CMAF) or gradient-based attribution from image tokens to textual tokens. In LLaVA-1.5-7B, as visualized in their Figure 20, these curves show that visual signals begin to noticeably influence textual representations

around the 10th decoder layer, reaching their strongest coupling between layers 12 and 14. The early layers (before layer 10) mainly encode low-level visual features without linguistic interaction, while the later layers (after layer 15) are dominated by text-only reasoning and exhibit minimal cross-modal feedback.

4. **Determination of the fusion onset.** The point at which the cross-modal influence curve first rises significantly above its baseline is identified as the *start of cross-modal fusion*. In their results, this transition occurs around layer $\ell \approx 10$ for LLaVA-1.5-7B, marking the onset of the vision–language interaction phase. Subsequent layers (approximately 10–14) are characterized as the principal fusion zone where semantic alignment and attention blending are most active.

Our adoption. Building on this analysis, we set the starting detection layer for CleanSight to $\ell_s = 10$, which corresponds to the beginning of the cross-modal fusion stage in LLaVA-1.5-7B. We then define a short consecutive window $\mathcal{L}_{\text{det}} = \{10, 11, 12\}$, covering the most discriminative middle layers where attention manipulation by backdoor triggers is empirically strongest. This configuration is consistent with the observations from Zhang et al. [18].

General recommendation. For other LVLM architectures, we recommend following the same diagnostic approach: (1) perform a layer-wise attention ablation or attribution analysis on a small validation set, (2) identify the first layer where visual→textual influence or fusion intensity exhibits a sharp increase, and (3) select a compact contiguous range (typically 2–4 layers) starting from that layer as $\mathcal{L}_{\text{det}}$. This ensures that the detector operates precisely in the regime where cross-modal reasoning emerges, while avoiding both low-level vision-only and high-level text-only regions of the network.

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